Executive Summary — Telco Customer Churn Analysis

1. Project Overview

The objective of this project is to analyze customer churn in a telecommunications company, identify customer segments with higher churn rates, and uncover patterns that can help the business improve customer retention.

The analysis uses Python for data cleaning, manipulation, exploratory data analysis (EDA), and visualization. I examined customer demographics, tenure, contract types, internet services, supplementary services, and payment methods to understand how customer characteristics and service choices relate to churn.

The dataset contains **7,043 customer records and 21 columns**, covering demographic information, account details, subscribed services, billing information, and customer churn status.

Dataset summary

Metric	Value
Total customers	7,043
Customers who churned	1,869
Customers who stayed	5,174
Overall churn rate	26.54%
Customer retention rate	73.46%
Average customer tenure	32.37 months
Average monthly charges	$64.76

These figures establish the baseline for evaluating the company's customer retention performance.

2. Data Cleaning and Preparation

Before performing the analysis, I inspected the dataset's structure, data types, missing values, and duplicate customer records.

The main preparation steps included:

- **Data type conversion:** Converted the `TotalCharges` column from text to numeric format so that it could be analyzed quantitatively.
- **Missing-value treatment:** Replaced blank `TotalCharges` entries with zero, based on the assumption that the corresponding customers had zero tenure and had not yet accumulated charges.
- **Duplicate validation:** Checked the `customerID` column and found no duplicate customer records.
- **Category standardization:** Converted the `SeniorCitizen` column from binary values (0 and 1) to readable categories (No and Yes).
- **Data quality checks:** Confirmed that the resulting dataset contained no missing values after the preparation steps.

The cleaned dataset was then used to create charts comparing churn counts and distributions across different customer groups.

3. Key Findings from the Analysis

3.1 Overall Customer Churn

The overall churn chart shows that 1,869 out of 7,043 customers discontinued their services.

Customer status	Customers	Percentage
Churned	1,869	26.54%
Retained	5,174	73.46%
Total	**7,043**	**100.00%**

Approximately **one in every four customers has churned**. Although most customers remain with the company, losing more than a quarter of the customer base represents a substantial retention challenge.

This finding provides the baseline against which the churn rates of individual customer segments should be compared.

Business implication: The company should investigate the characteristics and experiences of customers who leave, then prioritize retention initiatives for segments with consistently higher churn rates.

3.2 Customer Demographics: Gender and Senior Citizens

Gender analysis

The gender-based chart compares the number of male and female customers who churned or remained with the company.

Gender	Total customers	Churned customers	Churn rate
Female	3,488	939	26.92%
Male	3,555	930	26.16%

The churn rates differ by less than one percentage point. Gender therefore does not appear to be a strong differentiator in this dataset.

Business implication: Retention strategies should not prioritize customers based on gender alone. More informative characteristics, such as contract type, tenure, and service experience, deserve greater attention.

Senior-citizen analysis

The senior-citizen comparison shows a more substantial difference in churn behavior.

Customer segment	Total customers	Churned customers	Churn rate
Senior citizens	1,142	476	41.68%
Non-senior citizens	5,901	1,393	23.61%

Senior citizens have a churn rate of approximately **41.68%**, compared with **23.61%** among non-senior customers. This is a difference of approximately 18.07 percentage points.

The senior-citizen churn rate is about 1.77 times the rate of non-senior customers.

Business implication: The company should investigate whether senior customers experience different support needs, pricing concerns, or service difficulties. Targeted assistance and clearer communication could be tested, but the charts alone do not establish why this segment churns more frequently.

3.3 Customer Tenure

The tenure histogram compares customer service duration with churn status.

The chart suggests that churn is concentrated among customers with relatively short tenure, especially those in the first few months of service. Customers who have maintained their subscriptions for longer periods appear more likely to remain with the company.

This pattern is important because early customer departure can prevent the business from recovering its acquisition costs and developing a longer-term customer relationship.

Business implication:

- Introduce a structured onboarding process for new customers.
- Collect customer feedback during the first one to three months.
- Monitor early complaints, billing confusion, and service issues.
- Identify newly acquired customers who may need additional support.

The current histogram uses customer counts rather than tenure-specific churn percentages. To quantify the risk more precisely, the company should calculate churn rates for tenure groups such as 0–6 months, 7–12 months, 13–24 months, and more than 24 months.

3.4 Contract Type

The contract analysis reveals one of the clearest differences between customer segments.

Contract type	Total customers	Churned customers	Churn rate
Month-to-month	3,875	1,655	42.71%
One year	1,473	166	11.27%
Two years	1,695	48	2.83%

Month-to-month customers have a churn rate of approximately **42.71%**, compared with 11.27% for one-year contracts and 2.83% for two-year contracts.

The month-to-month churn rate is approximately 15 times the two-year contract churn rate.

This is a strong association between contract type and customer retention. However, customers who choose longer contracts may already be more committed to the company, so the difference cannot automatically be attributed to contract duration alone.

Business implication:

- Test incentives for customers to move from month-to-month to annual contracts.
- Offer clear explanations of the savings and benefits associated with longer commitments.
- Identify month-to-month customers with high churn risk and offer relevant retention options.
- Measure whether contract incentives reduce churn enough to justify their cost.

The objective should be to improve retention without relying on excessive discounts that reduce profitability.

3.5 Internet Services

The internet-service chart shows a higher churn count among fiber-optic customers than among DSL customers or customers without internet service.

Internet service	Total customers	Churned customers	Churn rate
Fiber optic	3,096	1,297	41.89%
DSL	2,421	459	18.96%
No internet service	1,526	113	7.40%

The churn rate for fiber-optic customers is approximately 41.89%, more than twice the rate among DSL customers.

This difference warrants further investigation. The chart does not establish whether the pattern is associated with pricing, customer expectations, service reliability, competition, or other characteristics of fiber-optic subscribers.

Business implication: The company should examine fiber-optic customer complaints, satisfaction ratings, service interruptions, pricing, and cancellation reasons. Any corrective action should be based on evidence from those additional sources.

3.6 Additional Internet Features and Technical Support

The analysis also compares churn across supplementary services, including online security, online backup, device protection, technical support, streaming television, and streaming movies.

The most pronounced differences appear in online security and technical support.

Online security status	Total customers	Churned customers	Churn rate
No	3,498	1,461	41.77%
Yes	2,019	295	14.61%
No internet service	1,526	113	7.40%

Customers without online security have a churn rate of approximately 41.77%, compared with 14.61% among customers who have the service.

Technical support status	Total customers	Churned customers	Churn rate
No	3,473	1,446	41.64%
Yes	2,044	310	15.17%
No internet service	1,526	113	7.40%

Similarly, customers without technical support have a churn rate of 41.64%, compared with 15.17% among customers who have technical support.

The charts also suggest higher churn among customers without online backup and device protection. By comparison, the presence or absence of streaming services shows less pronounced differences in churn

counts.

These findings indicate that supplementary services may be useful indicators for identifying customers who need closer attention. They do not prove that purchasing these services directly prevents churn, since other customer characteristics may influence both service adoption and retention.

Business implication:

- Investigate whether customers understand the benefits of supplementary services.
- Offer relevant service bundles rather than indiscriminate upselling.
- Review technical-support accessibility and customer satisfaction.
- Test whether improved service packages lead to measurable retention gains.

3.7 Payment Methods

The payment-method chart highlights electronic checks as the category with the largest number of churned customers.

Payment method	Total customers	Churned customers	Churn rate
Electronic check	2,365	1,071	45.31%
Mailed check	1,612	308	19.11%
Bank transfer (automatic)	1,544	258	16.71%
Credit card (automatic)	1,522	232	15.24%

Electronic-check customers have a churn rate of approximately **45.31%**, substantially higher than the rates for other payment methods.

By contrast, customers paying through automatic bank transfers or credit cards have churn rates of approximately 16.71% and 15.24%, respectively.

The difference makes payment method a useful variable for further investigation. However, the analysis does not establish that electronic checks themselves cause customers to leave. Payment method may be associated with other customer characteristics or preferences.

Business implication:

- Examine billing-related complaints and payment failures.
- Investigate whether electronic-check customers differ in contract type, tenure, or monthly charges.
- Consider communicating the convenience of automatic payments without forcing customers to change their preferred payment method.
- Measure churn rates before and after any payment-related intervention.

4. Consolidated Findings

The following table summarizes the main patterns identified in the analysis.

Factor	Higher-churn segment	Churn rate	Comparison
Overall churn	All customers	26.54%	Baseline

Factor	Higher-churn segment	Churn rate	Comparison
Senior-citizen status	Senior citizens	41.68%	23.61% for non-senior customers
Contract type	Month-to-month	42.71%	2.83% for two-year contracts
Internet service	Fiber optic	41.89%	18.96% for DSL
Online security	No online security	41.77%	14.61% for customers with security
Technical support	No technical support	41.64%	15.17% for customers with support
Payment method	Electronic check	45.31%	15.24% for automatic credit-card payments
Gender	Female	26.92%	26.16% for male customers

Churn rates are calculated as churned customers divided by all customers within the relevant segment. Percentages are rounded to two decimal places.

5. Business Recommendations and Priorities

Based on the patterns identified, the following actions should be considered.

Priority 1 — Improve early customer retention

Develop onboarding communications, early satisfaction checks, and proactive support for customers in their first months of service. Track churn by tenure group to identify the period with the greatest risk.

Priority 2 — Address month-to-month churn

Test targeted contract incentives and retention offers. Compare the cost of these incentives with the value of the customers retained.

Priority 3 — Investigate fiber-optic churn

Review service quality, pricing, complaints, and customer satisfaction among fiber-optic users before making changes to service delivery or pricing.

Priority 4 — Investigate payment-related churn

Examine whether electronic-check customers experience specific billing or payment issues. Test alternative payment communications and evaluate their effects on churn.

Priority 5 — Improve access to supplementary services

Investigate whether online security, backup, device protection, and technical support address customer needs. Test targeted bundles and measure both retention and profitability.

Priority 6 — Provide targeted customer support

Investigate the reasons for higher churn among senior customers and determine whether tailored communication or support would improve their experience.

6. Limitations and Further Analysis

The analysis is exploratory and identifies associations rather than causal relationships. Several improvements would make the findings more robust:

- **Segment-level validation:** Calculate churn rates for all categories instead of relying only on count charts.
- **Statistical testing:** Test whether differences in churn rates are statistically meaningful.
- **Multivariable analysis:** Examine whether factors such as contract type, tenure, internet service, and payment method remain associated with churn when analyzed together.
- **Predictive modeling:** Develop a classification model to identify customers at elevated risk of churn.
- **Financial impact analysis:** Estimate the revenue and customer lifetime value at risk and compare them with the cost of retention campaigns.
- **Intervention measurement:** Use controlled experiments or appropriate comparisons to determine whether retention actions actually reduce churn.

These steps would help distinguish the strongest independent predictors from factors that merely appear related to churn.

7. Conclusion

This project demonstrates how exploratory data analysis and visualization can transform a telecommunications customer dataset into actionable business insights.

The analysis identifies a baseline churn rate of 26.54% and highlights several segments with substantially higher churn, including month-to-month customers, fiber-optic subscribers, customers without online security or technical support, senior customers, and electronic-check users.

The most important next step is to validate these patterns using segment-level rates, statistical analysis, and predictive modeling. By prioritizing high-risk segments and measuring the results of retention initiatives, the company can develop a more evidence-based approach to customer retention.

Tools used: Python, Pandas, NumPy, Matplotlib, Seaborn, and exploratory data analysis.